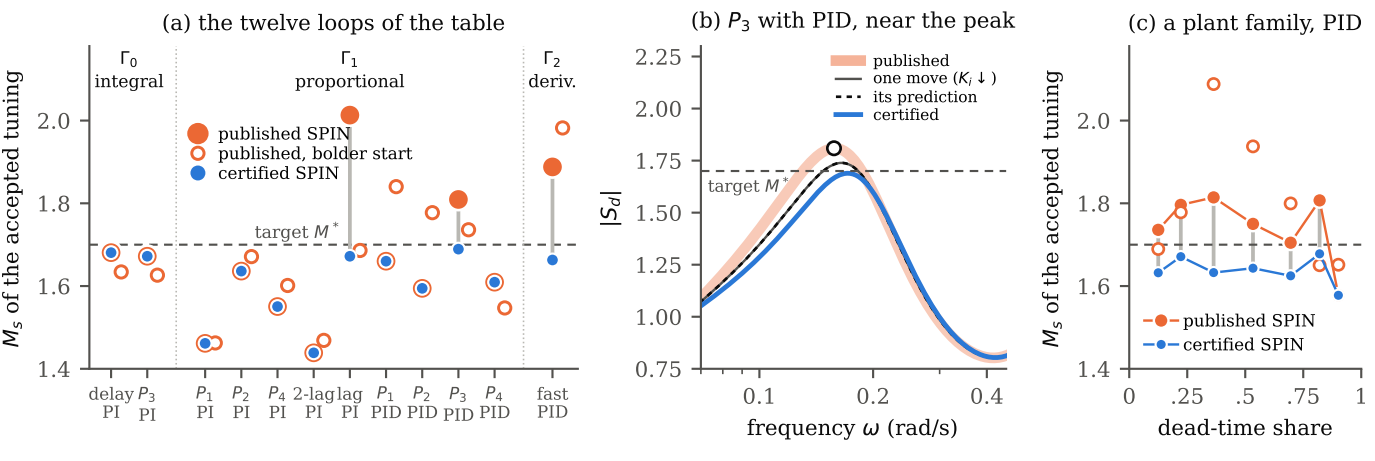
(a) the twelve loops of the table
Γ0 integral
Γ1 proportional
Γ2 deriv.
published SPIN
published, bolder start
certified SPIN
target M*
Ms of the accepted tuning
2.0
1.8
1.6
1.4
delay PI
P3 PI
P1 PI
P2 PI
P4 PI
2-lag PI
lag PI
P1 PID
P2 PID
P3 PID
P4 PID
fast PID
(b) P3 with PID, near the peak
published
one move (Ki ↓)
its prediction
certified
target M*
|Sd|
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.1
0.2
0.4
frequency ω (rad/s)
(c) a plant family, PID
Ms of the accepted tuning
2.0
1.8
1.6
1.4
published SPIN
certified SPIN
0
.25
.5
.75
1
dead-time share